

Functional Specification Document (FSD)

Project Title

Virtual Agent–Driven SLA Breach Awareness & Justification System

Version Information

Item	Details
Project Title	Virtual Agent–Driven SLA Breach Awareness & Justification System
Document Type	Functional Specification Document (FSD)
Version	1.0
Prepared By	Student Developer
Platform	ServiceNow, Flask
Status	Final

Table of Contents

- 1. Introduction
- 2. Purpose
- 3. Scope
- 4. System Overview
- 5. Objectives
- 6. Stakeholders
- 7. Functional Requirements
- 8. Non-Functional Requirements
- 9. User Roles
- 10. System Modules
- 11. Database Design
- 12. Workflow Description
- 13. Business Rules
- 14. Notifications
- 15. Security
- 16. Assumptions

- 17. Constraints
 - 18. Future Enhancements
 - 19. Conclusion
-

1. Introduction

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** is an intelligent IT Service Management (ITSM) application designed to improve SLA governance through proactive monitoring, workflow automation, and Artificial Intelligence. The application continuously monitors incident tickets, evaluates remaining SLA time, classifies incidents based on SLA risk, and assists support personnel using an AI-powered Virtual Agent.

The system integrates ServiceNow workflows, Google Gemini AI, Flask, and SQLite to automate SLA monitoring, generate intelligent breach justifications, provide corrective guidance, and deliver real-time dashboards that improve operational visibility and accountability.

2. Purpose

The purpose of this Functional Specification Document is to define the functional and technical requirements of the Virtual Agent–Driven SLA Breach Awareness & Justification System.

This document serves as the primary reference for developers, testers, administrators, and project stakeholders by describing the system's features, workflows, business rules, security requirements, and expected behavior.

3. Scope

The application provides an intelligent solution for monitoring Service Level Agreements throughout the incident lifecycle.

The project includes:

- User authentication
- Incident management
- Automatic SLA attachment
- Continuous SLA monitoring
- SLA risk classification
- Virtual Agent interaction
- Google Gemini AI integration
- Workflow automation
- Notification management

- Dashboard and reporting
- Role-Based Access Control (RBAC)

The application is intended for organizations that require proactive SLA governance and improved IT service management.

4. System Overview

The application consists of three primary user groups.

Support Agent

Support Agents can:

- View assigned incidents.
 - Monitor SLA status.
 - Receive SLA notifications.
 - Interact with the Virtual Agent.
 - Submit SLA breach justifications.
 - Update incident records.
-

Team Lead / Manager

Managers can:

- Monitor SLA compliance.
 - Review SLA breach reports.
 - Track team performance.
 - Analyze SLA trends.
 - View dashboards.
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Administrator

Administrators can:

- Manage user accounts.
- Configure workflows.
- Maintain dashboards.
- Generate reports.
- Configure system settings.

- Monitor application performance.
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5. Objectives

The application aims to achieve the following objectives:

- Monitor SLAs continuously.
 - Predict SLA breaches before deadlines are missed.
 - Automatically classify incidents by SLA risk.
 - Notify users proactively.
 - Generate AI-powered SLA breach explanations.
 - Improve accountability through mandatory acknowledgement and justification.
 - Automate SLA-related workflows.
 - Provide real-time dashboards.
 - Improve service quality.
 - Reduce SLA violations.
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6. Stakeholders

Stakeholder	Responsibility
Support Agent	Monitor incidents, respond to SLA alerts, update incidents
Team Lead / Manager	Review SLA compliance and team performance
Administrator	Manage users, workflows, dashboards, and reports
Project Guide	Review project implementation
SmartBridge Mentor	Evaluate project progress and documentation

7. Functional Requirements

The application shall provide the following functionality.

Authentication

- Secure user login.
- Session management.
- Role-based access.
- User authentication.

Incident Management

- Create incidents.
- Update incidents.
- Assign incidents.
- Track incident lifecycle.

SLA Monitoring

- Automatic SLA attachment.
- Remaining time calculation.
- Continuous monitoring.
- SLA threshold detection.

SLA Classification

Automatically classify incidents into:

- SAFE
- AT RISK
- BREACHED

Virtual Agent

- Display SLA information.
- Notify users.
- Explain SLA risks.
- Capture acknowledgement.
- Guide users during incident handling.

AI Integration

Google Gemini AI shall:

- Generate SLA breach justifications.
- Recommend corrective actions.
- Explain SLA risks.

- Assist support personnel.
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Workflow Automation

Flow Designer shall:

- Trigger notifications.
 - Execute workflows.
 - Automate SLA governance.
 - Update incident status.
-

Reporting

Generate:

- SLA compliance reports.
 - Incident reports.
 - Dashboard analytics.
 - Agent performance reports.
 - Breach trend analysis.
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8. Non-Functional Requirements

Performance

The system should:

- Monitor incidents in real time.
 - Generate fast AI responses.
 - Update dashboards efficiently.
 - Support multiple concurrent users.
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Reliability

The application should provide:

- Accurate SLA calculations.
- Stable workflow execution.
- Reliable AI integration.
- Consistent database operations.

Security

The application shall implement:

- Secure authentication.
- Role-Based Access Control (RBAC).
- Protected incident records.
- Secure data storage.

Scalability

The system should support:

- Increased incident volumes.
- Additional organizational users.
- Future AI enhancements.
- Enterprise deployment.

Availability

The application should remain operational during business hours with minimal downtime.

Usability

The application should provide:

- User-friendly interface.
- Easy navigation.
- Interactive dashboards.
- Simple Virtual Agent conversations.

9. User Roles

The Virtual Agent–Driven SLA Breach Awareness & Justification System supports three primary user roles, each with specific responsibilities and access permissions.

9.1 Support Agent

Support Agents are responsible for managing assigned incidents and ensuring SLA compliance.

Responsibilities

- Log into the application.
 - View assigned incidents.
 - Monitor SLA status.
 - Receive SLA alerts.
 - Interact with the Virtual Agent.
 - Acknowledge SLA risks.
 - Update incident information.
 - Submit SLA breach justifications.
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9.2 Team Lead / Manager

Managers supervise SLA compliance and monitor overall team performance.

Responsibilities

- Monitor SLA compliance.
 - View SLA dashboards.
 - Analyze SLA breach trends.
 - Review agent performance.
 - Monitor incident progress.
 - Review SLA justifications.
 - Generate SLA reports.
-

9.3 Administrator

Administrators manage the overall application and maintain system configuration.

Responsibilities

- Create and manage user accounts.
- Assign user roles.
- Configure workflows.
- Maintain dashboards.
- Generate reports.
- Configure system settings.
- Monitor application performance.
- Manage security settings.

10. System Modules

The application consists of several integrated modules that work together to provide intelligent SLA governance.

Module 1 – User Authentication

Description

Provides secure access to authorized users.

Features

- Secure login
- Session management
- User authentication
- Role validation

Input

- Username
- Password

Output

Authenticated user dashboard.

Module 2 – Incident Management

Description

Allows users to create, assign, update, and monitor incidents.

Features

- Create incident
- Update incident
- Assign incident
- View incident details
- Track incident history

Output

Updated incident records.

Module 3 – SLA Monitoring

Description

Continuously monitors the SLA associated with every incident.

Features

- Automatic SLA attachment
- Remaining SLA calculation
- SLA countdown timer
- Continuous monitoring
- Threshold detection

Output

Current SLA status.

Module 4 – SLA Risk Classification

Description

Automatically categorizes incidents based on the remaining SLA time.

Categories

- SAFE
- AT RISK
- BREACHED

Output

Updated SLA classification.

Module 5 – Virtual Agent

Description

Provides intelligent interaction between users and the system.

Features

- SLA notifications
- AI-generated guidance
- User acknowledgement
- Interactive conversations
- Incident assistance

Output

Updated incident workflow.

Module 6 – Google Gemini AI

Description

Generates intelligent SLA breach explanations and recommendations.

Features

- Context-aware responses
- Breach justification
- Corrective recommendations
- Decision support

Output

AI-generated SLA justification.

Module 7 – Workflow Automation

Description

Automates SLA-related activities using Flow Designer.

Features

- Notification generation
- Email alerts
- Workflow execution
- Incident updates
- SLA escalation

Output

Automated workflow completion.

Module 8 – Dashboard & Reporting

Description

Provides real-time operational insights.

Features

- SLA compliance dashboard

- Incident statistics
- Agent performance
- SLA trend analysis
- Breach reports

Output

Interactive dashboards and reports.

11. Database Design

The application stores operational information using SQLite.

User Table

Stores user account information.

Fields

- User ID
 - Name
 - Email
 - Username
 - Password
 - Role
-

Incident Table

Stores incident information.

Fields

- Incident ID
 - Incident Title
 - Description
 - Priority
 - Status
 - Assigned Agent
 - Created Date
-

SLA Table

Stores SLA information.

Fields

- SLA ID
 - Incident ID
 - SLA Start Time
 - SLA End Time
 - Remaining Time
 - SLA Status
 - Risk Classification
-

Justification Table

Stores AI-generated and user-submitted justifications.

Fields

- Justification ID
 - Incident ID
 - Generated Explanation
 - User Justification
 - Acknowledgement Status
 - Submitted Date
-

Notification Table

Stores notification history.

Fields

- Notification ID
 - Incident ID
 - Notification Type
 - Recipient
 - Status
 - Sent Time
-

Report Table

Stores dashboard and reporting data.

Fields

- Report ID
 - Incident Statistics
 - SLA Compliance
 - Breach Count
 - Performance Metrics
 - Generated Date
-

12. Workflow Description

The Virtual Agent–Driven SLA Breach Awareness & Justification System follows the workflow below.

Step 1

User logs into the application.



Step 2

A new incident is created or an existing incident is updated.



Step 3

The appropriate SLA is automatically attached to the incident.



Step 4

The SLA Monitoring Engine continuously tracks the remaining SLA time.



Step 5

The system automatically classifies the incident as:

- SAFE
- AT RISK
- BREACHED



Step 6

If an SLA threshold is reached, Flow Designer automatically triggers notifications.



Step 7

The Virtual Agent informs the assigned support agent about the SLA risk.



Step 8

Google Gemini AI analyzes the incident and generates an SLA breach explanation along with recommended corrective actions.



Step 9

The support agent acknowledges the SLA notification and submits the required justification.



Step 10

Incident records, dashboards, and reports are updated automatically.



Step 11

Managers monitor SLA compliance through dashboards and analytical reports.

13. Business Rules

The Virtual Agent–Driven SLA Breach Awareness & Justification System follows a set of business rules to ensure consistency, security, and effective SLA governance.

Authentication Rules

- Every user must log in using valid credentials.
- Access to application features is determined by the assigned user role.
- Unauthorized users cannot access restricted modules.
- User sessions are securely maintained throughout the application.

Incident Management Rules

- Every incident shall have a unique Incident ID.
 - Each incident must be assigned to a support agent.
 - Incident information can only be modified by authorized users.
 - Incident history shall be maintained for audit purposes.
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SLA Monitoring Rules

- Every incident must have an associated SLA.
 - SLA monitoring begins immediately after an incident is created.
 - Remaining SLA time shall be updated continuously.
 - SLA status shall be recalculated automatically whenever incident data changes.
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SLA Classification Rules

Incidents shall be automatically classified as:

- SAFE
- AT RISK
- BREACHED

The classification depends on the remaining SLA time and predefined SLA thresholds.

Virtual Agent Rules

- The Virtual Agent shall notify users when an incident becomes AT RISK.
 - Users shall acknowledge SLA warnings before proceeding.
 - AI-generated guidance shall be displayed during Virtual Agent conversations.
 - Virtual Agent interactions shall be recorded for future reference.
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AI Justification Rules

- Google Gemini AI shall generate SLA breach explanations based on incident details.
 - Users may review AI-generated recommendations before submitting updates.
 - AI responses shall support, but not replace, user decision-making.
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Reporting Rules

- Dashboards shall display the latest available incident data.
 - Reports shall include accurate SLA statistics.
 - Only authorized users may access administrative reports.
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14. Notifications

The application automatically generates notifications during important SLA events.

Support Agent Notifications

- Incident assigned.
 - SLA approaching warning threshold.
 - SLA classified as AT RISK.
 - SLA BREACHED.
 - AI-generated recommendation available.
 - Justification required.
 - Workflow completed.
-

Team Lead / Manager Notifications

- Multiple SLA breaches detected.
 - Team SLA compliance updates.
 - Daily SLA performance summary.
 - Escalated incidents requiring attention.
-

Administrator Notifications

- Workflow execution status.
 - User management updates.
 - System activity alerts.
 - Dashboard generation status.
-

Notification Channels

Notifications may be delivered through:

- Virtual Agent
 - ServiceNow Service Portal
 - Email Notifications
-

15. Security

Security is implemented using industry-standard authentication and authorization mechanisms.

Authentication

The application provides:

- Secure login
 - Credential verification
 - Session management
 - User authentication
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Authorization

Role-Based Access Control (RBAC) restricts access based on user roles.

Support Agent

Permissions:

- View assigned incidents
 - Update incidents
 - Submit SLA justifications
 - Interact with Virtual Agent
-

Team Lead / Manager

Permissions:

- View dashboards
 - Monitor SLA compliance
 - Generate reports
 - Review justifications
-

Administrator

Permissions:

- Full application access
 - User management
 - Workflow configuration
 - Dashboard management
 - Security administration
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Data Protection

The application ensures:

- Secure database storage.
 - Controlled user permissions.
 - Protection of incident information.
 - Secure handling of AI-generated content.
 - Audit logging for important system activities.
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16. Assumptions

The following assumptions were considered during system development:

- Every user has valid login credentials.
 - Every incident has an associated SLA.
 - ServiceNow services remain operational.
 - Google Gemini AI services are available.
 - Users have a stable internet connection.
 - Email notification services are configured correctly.
 - Support agents acknowledge SLA warnings promptly.
 - Incident information is entered accurately.
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17. Constraints

The following constraints apply to the application:

- Internet connectivity is required.
 - Google Gemini AI depends on external API availability.
 - Workflow execution depends on ServiceNow services.
 - SQLite is intended for prototype and academic implementations.
 - Dashboard accuracy depends on current incident information.
 - Application performance may vary under extremely large workloads.
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18. Future Enhancements

The Virtual Agent–Driven SLA Breach Awareness & Justification System can be enhanced through additional features.

Possible future enhancements include:

- Machine Learning models for predictive SLA forecasting.

- Mobile application support.
- Integration with Microsoft Teams and Slack.
- SMS and push notification services.
- Voice-enabled Virtual Agent interactions.
- Cloud database integration.
- Multi-language Virtual Agent support.
- Advanced Business Intelligence dashboards.
- Automatic incident prioritization using AI.
- Integration with enterprise ITSM platforms.
- Predictive workload balancing for support teams.
- Real-time anomaly detection in SLA performance.

These enhancements will improve scalability, intelligence, and enterprise adoption.

19. Conclusion

The Functional Specification Document defines the functional behavior and operational requirements of the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. It describes the application's architecture, workflows, user roles, business rules, security mechanisms, notification processes, and system constraints.

The application combines proactive SLA monitoring, automated workflow execution, Google Gemini AI, Virtual Agent interaction, and real-time analytics to improve IT Service Management. By predicting SLA risks before deadlines are missed and providing intelligent recommendations, the system enables organizations to reduce SLA breaches, strengthen accountability, improve operational efficiency, and enhance service quality.

This Functional Specification Document serves as the primary reference for development, testing, deployment, maintenance, and future enhancement of the application while ensuring that the project objectives are successfully achieved